experimentation

March 12, 2026

1 1. Data Preprocessing

1.1 Converting provided PDF into individual page Images

[1]:
```
from src.dataUtils import pdf_to_images, transcript_to_page_txt
```

[3]:
```
######### Handwritten Pages ##########
pdf_to_images(
    pdf_path="data/1.raw/HandwrittenPages/AHPG-GPAH AU61&#x3a;2 - 1606.pdf",
    output_dir="data/2.images/HandwrittenPages",
    page_range=(1,1)
)
pdf_to_images(
    pdf_path="data/1.raw/HandwrittenPages/PT3279&#x3a;146&#x3a;342 - 1857.pdf",
    output_dir="data/2.images/HandwrittenPages",
    page_range=(1,1)
)
pdf_to_images(
    pdf_path="data/1.raw/HandwrittenPages/AHPG-GPAH 1&#x3a;1716,A.35 - 1744.
 ↪pdf",
    output_dir="data/2.images/HandwrittenPages",
    page_range=(1,1)
)
pdf_to_images(
    pdf_path="data/1.raw/HandwrittenPages/ES.28079.AHN&#x3a;&#x3a;
 ↪INQUISICIÓN,1667,Exp.12 - 1640.pdf",
    output_dir="data/2.images/HandwrittenPages",
    page_range=(1,1)
)
pdf_to_images(
    pdf_path="data/1.raw/HandwrittenPages/Pleito entre el Marqués de Viana.pdf",
    output_dir="data/2.images/HandwrittenPages",
    page_range=(1,1)
)
########## Printing Pages ##########
pdf_to_images(
    pdf_path="data/1.raw/PrintedPages/PORCONES.23.5 - 1628.pdf",
    output_dir="data/2.images/PrintedPages",
```

```
    page_range=(1,4)
)
pdf_to_images(
    pdf_path="data/1.raw/PrintedPages/Guardiola - Tratado nobleza.pdf",
    output_dir="data/2.images/PrintedPages",
    page_range=(12,14)
)
pdf_to_images(
    pdf_path="data/1.raw/PrintedPages/PORCONES.748.6 - 1650.pdf",
    output_dir="data/2.images/PrintedPages",
    page_range=(1,4)
)
pdf_to_images(
    pdf_path="data/1.raw/PrintedPages/Buendia - Instruccion.pdf",
    output_dir="data/2.images/PrintedPages",
    page_range=(2,4)
)
pdf_to_images(
    pdf_path="data/1.raw/PrintedPages/Covarrubias - Tesoro lengua.pdf",
    output_dir="data/2.images/PrintedPages",
    page_range=(7,9)
)
pdf_to_images(
    pdf_path="data/1.raw/PrintedPages/PORCONES.228.38 - 1646.pdf",
    output_dir="data/2.images/PrintedPages",
    page_range=(1,5)
)
```

```
Converting AHPG-GPAH AU61&#x3a;2 - 1606: 100%|      | 1/1 [00:03<00:00,
3.21s/it]

1 images written → data/2.images/HandwrittenPages/AHPG-GPAH AU61&#x3a;2 - 1606

Converting PT3279&#x3a;146&#x3a;342 - 1857: 100%|      | 1/1 [00:02<00:00,
2.92s/it]

1 images written → data/2.images/HandwrittenPages/PT3279&#x3a;146&#x3a;342 -
1857

Converting AHPG-GPAH 1&#x3a;1716,A.35 - 1744: 100%|      | 1/1 [00:03<00:00,
3.25s/it]

1 images written → data/2.images/HandwrittenPages/AHPG-GPAH 1&#x3a;1716,A.35 -
1744

Converting ES.28079.AHN&#x3a;&#x3a;INQUISICIÓN,1667,Exp.12 - 1640:
100%|      | 1/1 [00:00<00:00,  6.02it/s]

1 images written →
data/2.images/HandwrittenPages/ES.28079.AHN&#x3a;&#x3a;INQUISICIÓN,1667,Exp.12 -
1640
```

```
Converting Pleito entre el Marqués de Viana: 100%|      | 1/1 [00:00<00:00,
4.86it/s]

1 images written → data/2.images/HandwrittenPages/Pleito entre el Marqués de
Viana

Converting PORCONES.23.5 - 1628: 100%|      | 4/4 [00:11<00:00,  2.97s/it]

7 images written → data/2.images/PrintedPages/PORCONES.23.5 - 1628

Converting Guardiola - Tratado nobleza: 100%|      | 3/3 [00:00<00:00,
5.94it/s]

3 images written → data/2.images/PrintedPages/Guardiola - Tratado nobleza

Converting PORCONES.748.6 - 1650: 100%|      | 4/4 [00:11<00:00,  2.84s/it]

4 images written → data/2.images/PrintedPages/PORCONES.748.6 - 1650

Converting Buendia - Instruccion: 100%|      | 3/3 [00:00<00:00,  5.55it/s]

6 images written → data/2.images/PrintedPages/Buendia - Instruccion

Converting Covarrubias - Tesoro lengua: 100%|      | 3/3 [00:00<00:00,
3.03it/s]

3 images written → data/2.images/PrintedPages/Covarrubias - Tesoro lengua

Converting PORCONES.228.38 - 1646: 100%|      | 5/5 [00:14<00:00,  2.82s/it]

5 images written → data/2.images/PrintedPages/PORCONES.228.38 - 1646
```

1.2 Converting provided transcripts to individual txt pages

[4]:
```
transcript_to_page_txt(
    input_path="data/transcripts/HandwrittenPages/AHPG-GPAH 1.1716,A.35 -␣
↪1744_transcription.docx",
    output_dir="data/transcripts/HandwrittenPages/")
transcript_to_page_txt(
    input_path="data/transcripts/HandwrittenPages/AHPG-GPAH AU61.2 -␣
↪1606_transcription.docx",
    output_dir="data/transcripts/HandwrittenPages/")
transcript_to_page_txt(
    input_path="data/transcripts/HandwrittenPages/ES.28079.AHN.
↪INQUISICIÓN,1667,Exp.12 - 1640_transcription.docx",
    output_dir="data/transcripts/HandwrittenPages/")
transcript_to_page_txt(
    input_path="data/transcripts/HandwrittenPages/Pleito entre el Marqués de␣
↪Viana_transcription.docx",
    output_dir="data/transcripts/HandwrittenPages/")
transcript_to_page_txt(
```

```
    input_path="data/transcripts/HandwrittenPages/PT3279.146.342 -␣
↪1857_transcription.docx",
    output_dir="data/transcripts/HandwrittenPages/")
transcript_to_page_txt(
    input_path="data/transcripts/PrintedPages/Buendia - Instruccion␣
↪transcription.docx",
    output_dir="data/transcripts/PrintedPages/")
transcript_to_page_txt(
    input_path="data/transcripts/PrintedPages/Covarrubias - Tesoro lengua␣
↪transcription.docx",
    output_dir="data/transcripts/PrintedPages/")
transcript_to_page_txt(
    input_path="data/transcripts/PrintedPages/Guardiola - Tratado nobleza␣
↪transcription.docx",
    output_dir="data/transcripts/PrintedPages/")
transcript_to_page_txt(
    input_path="data/transcripts/PrintedPages/PORCONES.23.5 - 1628␣
↪transcription.docx",
    output_dir="data/transcripts/PrintedPages/")
transcript_to_page_txt(
    input_path="data/transcripts/PrintedPages/PORCONES.228.38 - 1646␣
↪transcription.docx",
    output_dir="data/transcripts/PrintedPages/")
transcript_to_page_txt(
    input_path="data/transcripts/PrintedPages/PORCONES.748.6 - 1650␣
↪Transcription.docx",
    output_dir="data/transcripts/PrintedPages/")
```

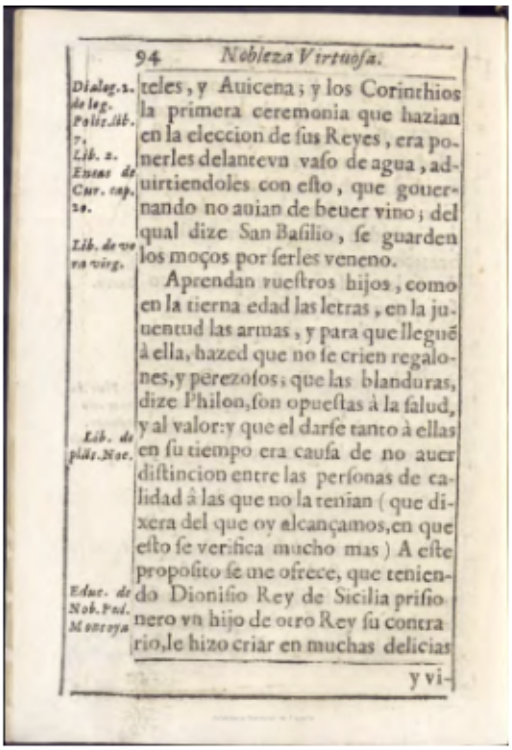
```
Creating directory → data/transcripts/HandwrittenPages/AHPG-GPAH 1.1716,A.35 -
1744_transcription

Parsing AHPG-GPAH 1.1716,A.35 - 1744_transcription: 100%|      | 28/28
[00:00<00:00, 343393.31it/s]
Writing AHPG-GPAH 1.1716,A.35 - 1744_transcription: 100%|      | 1/1
[00:00<00:00, 5817.34it/s]

1 pages written → data/transcripts/HandwrittenPages/AHPG-GPAH 1.1716,A.35 -
1744_transcription
Creating directory → data/transcripts/HandwrittenPages/AHPG-GPAH AU61.2 -
1606_transcription

Parsing AHPG-GPAH AU61.2 - 1606_transcription: 100%|      | 32/32
[00:00<00:00, 528416.25it/s]
Writing AHPG-GPAH AU61.2 - 1606_transcription: 100%|      | 1/1
[00:00<00:00, 5809.29it/s]

1 pages written → data/transcripts/HandwrittenPages/AHPG-GPAH AU61.2 -
1606_transcription
Creating directory →
```

```
data/transcripts/HandwrittenPages/ES.28079.AHN.INQUISICIÓN,1667,Exp.12 -
1640_transcription

Parsing ES.28079.AHN.INQUISICIÓN,1667,Exp.12 - 1640_transcription:
100%|      | 23/23 [00:00<00:00, 344532.11it/s]
Writing ES.28079.AHN.INQUISICIÓN,1667,Exp.12 - 1640_transcription:
100%|      | 1/1 [00:00<00:00, 5825.42it/s]

1 pages written → 
data/transcripts/HandwrittenPages/ES.28079.AHN.INQUISICIÓN,1667,Exp.12 -
1640_transcription
Creating directory → data/transcripts/HandwrittenPages/Pleito entre el Marqués
de Viana_transcription

Parsing Pleito entre el Marqués de Viana_transcription: 100%|      | 31/31
[00:00<00:00, 565319.23it/s]
Writing Pleito entre el Marqués de Viana_transcription: 100%|      | 1/1
[00:00<00:00, 5714.31it/s]

1 pages written → data/transcripts/HandwrittenPages/Pleito entre el Marqués de
Viana_transcription
Creating directory → data/transcripts/HandwrittenPages/PT3279.146.342 -
1857_transcription

Parsing PT3279.146.342 - 1857_transcription: 100%|      | 30/30
[00:00<00:00, 542366.90it/s]
Writing PT3279.146.342 - 1857_transcription: 100%|      | 1/1 [00:00<00:00,
6260.16it/s]

1 pages written → data/transcripts/HandwrittenPages/PT3279.146.342 -
1857_transcription
Creating directory → data/transcripts/PrintedPages/Buendia - Instruccion
transcription

Parsing Buendia - Instruccion transcription: 100%|      | 10/10
[00:00<00:00, 172605.10it/s]
Writing Buendia - Instruccion transcription: 100%|      | 3/3 [00:00<00:00,
13231.24it/s]

3 pages written → data/transcripts/PrintedPages/Buendia - Instruccion
transcription
Creating directory → data/transcripts/PrintedPages/Covarrubias - Tesoro lengua
transcription

Parsing Covarrubias - Tesoro lengua transcription: 100%|      | 9/9
[00:00<00:00, 106035.78it/s]
Writing Covarrubias - Tesoro lengua transcription: 100%|      | 3/3
[00:00<00:00, 11194.76it/s]

3 pages written → data/transcripts/PrintedPages/Covarrubias - Tesoro lengua
transcription
Creating directory → data/transcripts/PrintedPages/Guardiola - Tratado nobleza
transcription
```

```
Parsing Guardiola - Tratado nobleza transcription: 100%|      | 7/7
[00:00<00:00, 132252.83it/s]
Writing Guardiola - Tratado nobleza transcription: 100%|      | 3/3
[00:00<00:00, 12985.46it/s]

3 pages written → data/transcripts/PrintedPages/Guardiola - Tratado nobleza
transcription
Creating directory → data/transcripts/PrintedPages/PORCONES.23.5 - 1628
transcription

Parsing PORCONES.23.5 - 1628 transcription:  99%|      | 205/207
[00:00<00:00, 1270062.51it/s]
Writing PORCONES.23.5 - 1628 transcription: 100%|      | 5/5 [00:00<00:00,
8694.66it/s]

5 pages written → data/transcripts/PrintedPages/PORCONES.23.5 - 1628
transcription
Creating directory → data/transcripts/PrintedPages/PORCONES.228.38 - 1646
transcription

Parsing PORCONES.228.38 - 1646 transcription: 100%|      | 199/199
[00:00<00:00, 863150.46it/s]
Writing PORCONES.228.38 - 1646 transcription: 100%|      | 5/5 [00:00<00:00,
13751.82it/s]

5 pages written → data/transcripts/PrintedPages/PORCONES.228.38 - 1646
transcription
Creating directory → data/transcripts/PrintedPages/PORCONES.748.6 - 1650
Transcription

Parsing PORCONES.748.6 - 1650 Transcription:  98%|      | 134/137
[00:00<00:00, 1178274.08it/s]
Writing PORCONES.748.6 - 1650 Transcription: 100%|      | 4/4 [00:00<00:00,
14513.16it/s]

4 pages written → data/transcripts/PrintedPages/PORCONES.748.6 - 1650
Transcription
```

[13]:
```python
from src.dataUtils import display_sample
display_sample("data/2.images/PrintedPages/Padilla - 1 Nobleza virtuosa (1637)/
↪page111.png")
```

Sample Image: page111.png



94 Nobleza Virtuosa.

Dialog. 2. de leg. Polit. lib. 7. teles, y Auicena; y los Corinthios la primera ceremonia que hazian en la eleccion de sus Reyes, era ponerles delante vn vaso de agua, aduirtiendoles con esto, que gouernando no auian de beuer vino; del qual dize San Basilio, se guarden los moços por serles veneno.

Lib. 2. Enseñ. de Cur. cap. 20.

Lib. de vera virg.

Aprendan vuestros hijos, como en la tierna edad las letras, en la juuentud las armas, y para que lleguẽ à ella, hazed que no se crien regalones, y perezosos; que las blanduras, dize Philon, son opuestas à la salud, y al valor: y que el darse tanto à ellas en su tiempo era causa de no auer distincion entre las personas de calidad à las que no la tenian (que dixera del que oy alcançamos, en que esto se verifica mucho mas) A este proposito se me ofrece, que teniendo Dionisio Rey de Sicilia prisionero vn hijo de otro Rey su contrario, le hizo criar en muchas delicias

Lib. de plás. Noc.

Educ. de Nob. Pud. Montoya

y vi-

1.3 Preprocessing Images

Various operations:

```
OP_REGISTRY = {
    "grayscale": convert_to_grayscale,
    "deskew": correct_skew,
    "normalize": normalize_image,
    "denoise": denoise_image,
    "contrast": enhance_contrast,
    "binarize": binarize_image,
    "morph": morphological_operations
}
```

sample pipeline

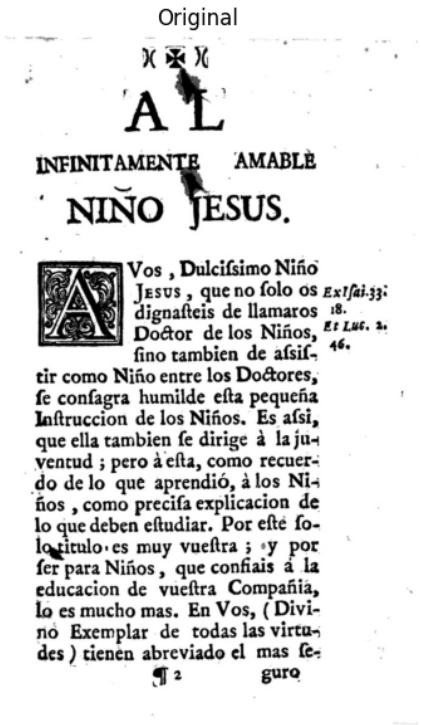
```
# default pipeline that applies to all pages except if overridden by page_pipelines(for specif
default_pipeline = [
     {"op": "grayscale"},
     {"op": "deskew"},
     {"op": "binarize"}
]
 page_pipelines = {
```

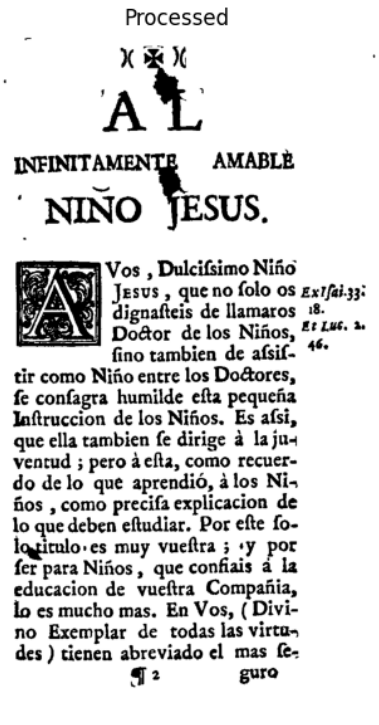
```
    (2,3): [
        {"op": "contrast"},
    ],

    8: [
        {"op": "deskew"},
        {"op": "binarize", "params": {"method":"adaptive"}}
    ]
}
```

[1]:
```
from src.dataUtils import process_image_folder, compare_images
```

[2]:
```
book1_pipeline = [
    {"op": "grayscale"},
    {"op": "deskew"},
    {"op": "binarize", "params": {"method": "otsu"}},

    #   Blob / noise cleaning (after binarization)
    {"op": "remove_noise", "params": {"max_area": 15}},      # remove speckles
    {"op": "remove_blobs", "params": {                       # remove large ink␣
 ↪blobs
        "min_area": 2000,
        "min_solidity": 0.55,
        "max_aspect_ratio": 4.0,
        "erosion_ratio": 0.35
    }},
    {"op": "morph_open",   "params": {"kernel_size": 3}},    # break thin␣
 ↪artifacts
]
process_image_folder(
    input_dir="data/2.images/PrintedPages/Buendia - Instruccion",
    output_dir="data/3.processed",
    default_pipeline=book1_pipeline
)
compare_images(
    original_path="data/2.images/PrintedPages/Buendia - Instruccion/page2_right.
 ↪png",
    processed_path="data/3.processed/Buendia - Instruccion/page2_right.png"
)
```

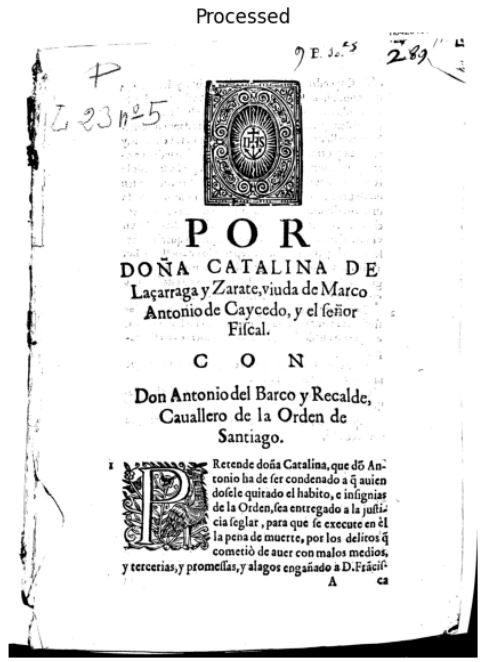
```
Processing Buendia - Instruccion: 100%|      | 6/6 [00:01<00:00,
3.20page/s]
```





[2]:
```
book2_pipeline = [
     {"op": "grayscale"},
     {"op": "deskew"},
     {"op": "normalize"},
     {"op": "denoise","params": {"method" : "nlm"}},
      {"op" : "morph","params" : {"operation" : "open", "k" : (2,2),␣
 ↪"iterations" : 2}},
     {"op": "binarize","params": {"method" : "otsu"}}
]
process_image_folder(
    input_dir="data/2.images/PrintedPages/PORCONES.23.5 - 1628",
    output_dir="data/3.processed",
    default_pipeline=book2_pipeline
)
compare_images(
    original_path="data/2.images/PrintedPages/PORCONES.23.5 - 1628/page1.png",
    processed_path="data/3.processed/PORCONES.23.5 - 1628/page1.png"
)
```

```
Processing PORCONES.23.5 - 1628: 100%|      | 7/7 [00:07<00:00,  1.14s/page]
```



1.4 Layout Detection

Using paddle ocr model for layout detection. As the model was giving OOM error so I resized the images to low dimension(based on aspect ratio) to perform layout analysis(this also improved speed). The the layout bboxes were rescaled based on the ratio we used to resize the image.

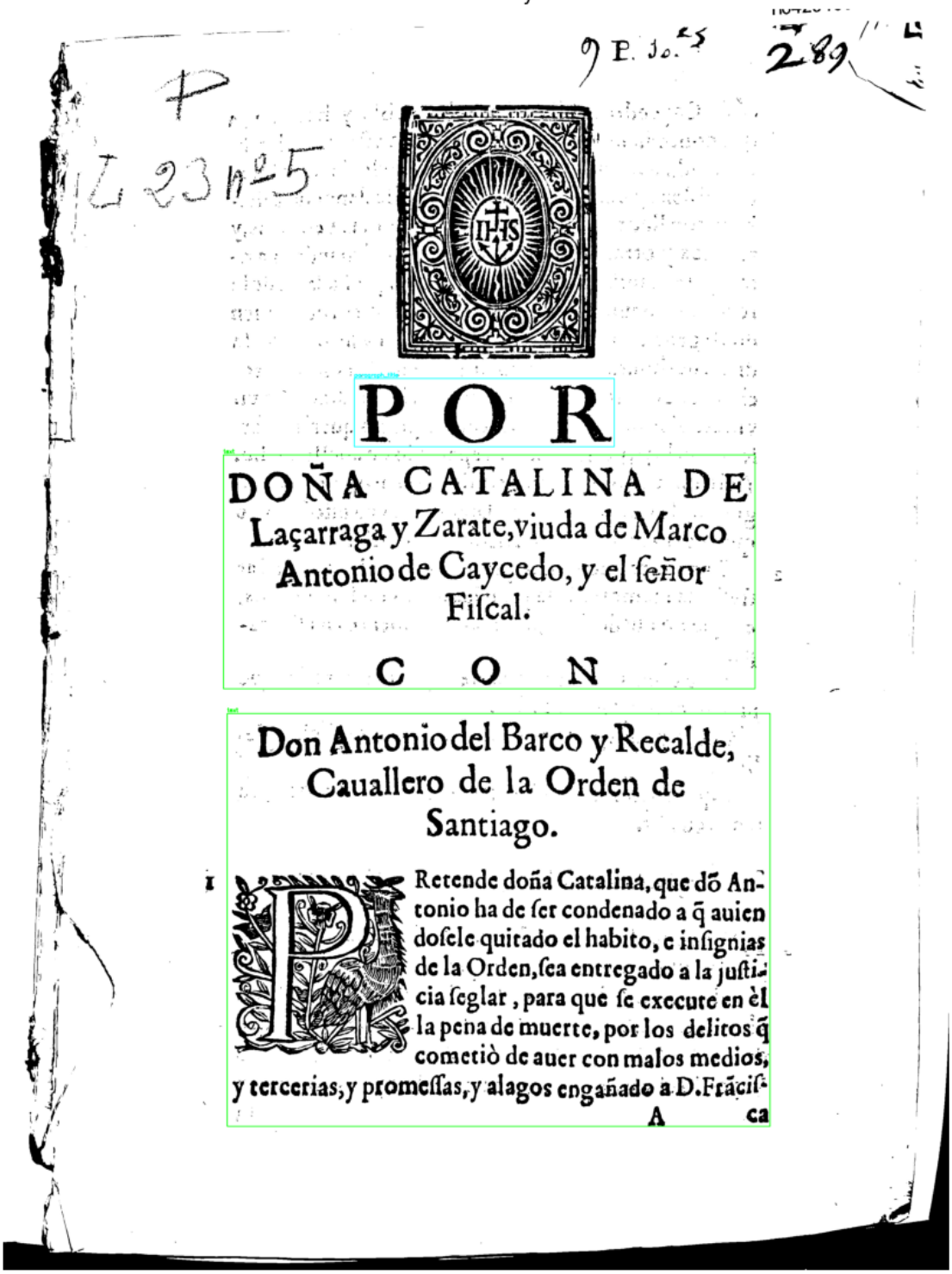
[1]:
```
from src.textDetection import detect_layout
import cv2

img = cv2.imread("data/3.processed/PORCONES.23.5 - 1628/page1.png")

layout = detect_layout(
    img,
    layout_model="PP-DocLayout_plus-L",
    visualize=True,
    max_layout_side=1500
)
```

```
/home/sarthak/miniconda3/envs/RenaissanceDetect/lib/python3.10/site-
packages/paddle/utils/cpp_extension/extension_utils.py:712: UserWarning: No
ccache found. Please be aware that recompiling all source files may be required.
You can download and install ccache from:
https://github.com/ccache/ccache/blob/master/doc/INSTALL.md
  warnings.warn(warning_message)
/home/sarthak/miniconda3/envs/RenaissanceDetect/lib/python3.10/site-
packages/requests/__init__.py:113: RequestsDependencyWarning: urllib3 (2.6.3) or
```

```
chardet (6.0.0.post1)/charset_normalizer (3.4.4) doesn't match a supported
version!
  warnings.warn(
Checking connectivity to the model hosters, this may take a while. To

bypass this check, set `PADDLE_PDX_DISABLE_MODEL_SOURCE_CHECK` to `True`.
Creating model: ('PP-DocLayout_plus-L', None)
Model files already exist. Using cached files. To redownload, please delete

the directory manually: `/home/sarthak/.paddlex/official_models/PP-

DocLayout_plus-L`.
Creating model: ('PP-LCNet_x1_0_textline_ori', None)
Model files already exist. Using cached files. To redownload, please delete

the directory manually: `/home/sarthak/.paddlex/official_models/PP-

LCNet_x1_0_textline_ori`.
Creating model: ('PP-OCRv5_server_det', None)
Model files already exist. Using cached files. To redownload, please delete

the directory manually: `/home/sarthak/.paddlex/official_models/PP-

OCRv5_server_det`.
Creating model: ('PP-OCRv5_server_rec', None)
Model files already exist. Using cached files. To redownload, please delete

the directory manually: `/home/sarthak/.paddlex/official_models/PP-

OCRv5_server_rec`.
Creating model: ('PP-Chart2Table', None)
Model files already exist. Using cached files. To redownload, please delete

the directory manually: `/home/sarthak/.paddlex/official_models/PP-

Chart2Table`.
Special tokens have been added in the vocabulary, make sure the associated

word embeddings are fine-tuned or trained.
Loading configuration file /home/sarthak/.paddlex/official_models/PP-

Chart2Table/config.json
Loading weights file /home/sarthak/.paddlex/official_models/PP-

Chart2Table/model_state.pdparams
Loaded weights file from disk, setting weights to model.
All model checkpoint weights were used when initializing

PPChart2TableInference.
```

```
All the weights of PPChart2TableInference were initialized from the model
checkpoint at /home/sarthak/.paddlex/official_models/PP-Chart2Table.
If your task is similar to the task the model of the checkpoint was trained on,
you can already use PPChart2TableInference for predictions without further
training.
Loading configuration file /home/sarthak/.paddlex/official_models/PP-
Chart2Table/generation_config.json
```

Text + Title Layout



POR

DOÑA CATALINA DE Laçarraga y Zarate, viuda de Marco Antonio de Caycedo, y el ſeñor Fiſcal.

CON

Don Antonio del Barco y Recalde, Cauallero de la Orden de Santiago.

PRetende doña Catalina, que dõ Antonio ha de ſer condenado a q̃ auiendoſele quitado el habito, e inſignias de la Orden, ſea entregado a la juſticia ſeglar, para que ſe execute en èl la pena de muerte, por los delitos q̃ cometiò de auer con malos medios, y tercerias, y promeſſas, y alagos engañado a D. Frãciſca

[3]:
```
layout
```

[3]:
```
[{'label': 'text',
  'bbox': [704, 1425, 2403, 2173],
  'score': 0.6307090520858765},
 {'label': 'text',
  'bbox': [716, 2252, 2451, 3572],
  'score': 0.6683927178382874},
 {'label': 'image',
  'bbox': [1259, 294, 1899, 1119],
  'score': 0.9162530303001404},
 {'label': 'image',
  'bbox': [672, 2739, 1297, 3357],
  'score': 0.8360862731933594},
 {'label': 'paragraph_title',
  'bbox': [1122, 1181, 1953, 1399],
  'score': 0.5278884172439575}]
```

2 2. Text Detection(Line Level)

[1]:
```python
import cv2
import numpy as np
import matplotlib.pyplot as plt
from paddleocr import PaddleOCR

img_path = "data/3.processed/PORCONES.23.5 - 1628/page1.png"
image = cv2.imread(img_path)
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)


# 1. Initialize the OCR engine
ocr = PaddleOCR(
    # Specify the exact official model names
    text_detection_model_name='PP-OCRv5_server_det',
    text_recognition_model_name='PP-OCRv5_server_rec',

    # Optional: Turn off the extra document processing modules to save RAM/Time
    use_doc_orientation_classify=False,
    use_doc_unwarping=False,
    use_textline_orientation=False,
    lang='sp'
)
```

```
/home/sarthak/miniconda3/envs/RenaissanceDetect/lib/python3.10/site-
packages/requests/__init__.py:113: RequestsDependencyWarning: urllib3 (2.6.3) or
chardet (6.0.0.post1)/charset_normalizer (3.4.4) doesn't match a supported
version!
  warnings.warn(
```

```
Checking connectivity to the model hosters, this may take a while. To
bypass this check, set `PADDLE_PDX_DISABLE_MODEL_SOURCE_CHECK` to `True`.
/tmp/ipykernel_1811208/716777847.py:12: UserWarning: `lang` and `ocr_version`
will be ignored when model names or model directories are not `None`.
  ocr = PaddleOCR(
Creating model: ('PP-OCRv5_server_det', None)
Model files already exist. Using cached files. To redownload, please delete
the directory manually: `/home/sarthak/.paddlex/official_models/PP-
OCRv5_server_det`.
Creating model: ('PP-OCRv5_server_rec', None)
Model files already exist. Using cached files. To redownload, please delete
the directory manually: `/home/sarthak/.paddlex/official_models/PP-
OCRv5_server_rec`.
```

2.0.1 Using only Text Detection Model

[2]:
```python
results = ocr.predict(img_path)

# from src.textDetection import merge_boxes_vertical_overlap
import numpy as np
from scipy.ndimage import gaussian_filter1d

vis = image_rgb.copy()
for res in results:
    bounding_boxes = res['dt_polys']
    # bounding_boxes = merge_boxes_vertical_overlap(res['dt_polys'])

    for box in bounding_boxes:
        pts = np.array(box, np.int32)
        pts = pts.reshape((-1, 1, 2))

        cv2.polylines(vis, [pts], isClosed=True, color=(0, 255, 0), thickness=2)

plt.figure(figsize=(12, 12))  # Adjust size as needed
plt.imshow(vis)
plt.axis('off')  # Hide the axes for a cleaner look
plt.title(f"Detected {len(bounding_boxes)} Text Regions", fontsize=16)
plt.show()
```

```
Resized image size (3024x4032) exceeds max_side_limit of 4000. Resizing to
fit within limit.
```

Detected 125 Text Regions

POR

DOÑA CATALINA DE Laçarraga y Zarate, viuda de Marco Antonio de Caycedo, y el ſeñor Fiſcal.

CON

Don Antonio del Barco y Recalde, Cauallero de la Orden de Santiago.

PRetende doña Catalina, que dõ Antonio ha de ſer condenado a q̃ auiendoſele quitado el habito, e inſignias de la Orden, ſea entregado a la juſticia ſeglar, para que ſe execute en èl la pena de muerte, por los delitos q̃ cometiò de auer con malos medios, y tercerias, y promeſſas, y alagos engañado a D. Frãciſca

2.0.2 Layout-aware Line Detection

Run PaddleOCR text detection **per layout region** (text / paragraph_title only), then offset each detected polygon back to full-page coordinates and merge into final line-level bounding boxes.

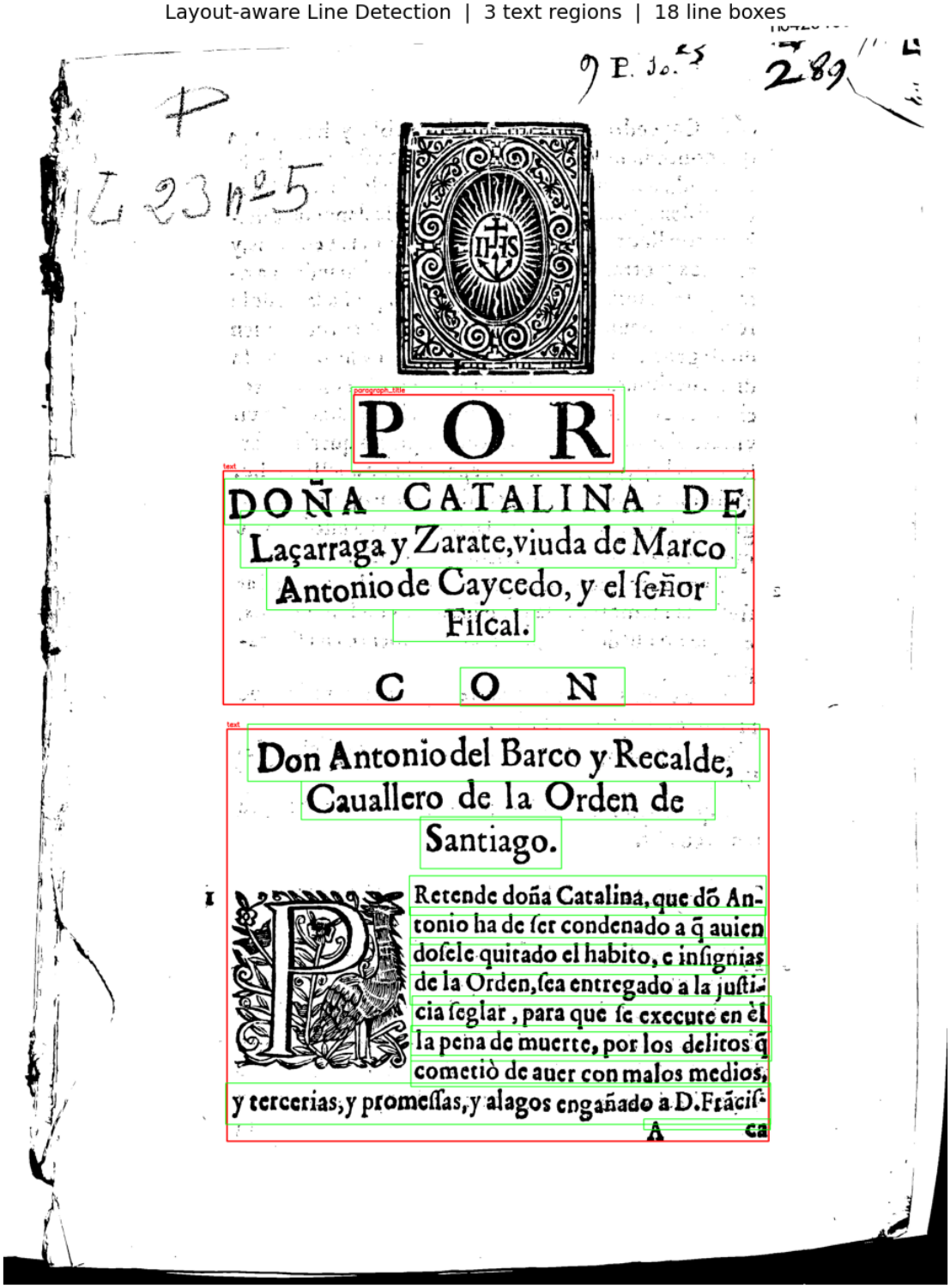
[5]:
```
import numpy as np
import cv2

#   Config
TEXT_LABELS       = {'text', 'paragraph_title', 'doc_title', 'header'}
REGION_PADDING    = 50     # white pixels added around every crop
LAYOUT_EXPAND     = 2     # slight outward expansion of each layout bbox
SCORE_THRESH      = 0.5    # discard raw OCR boxes below this confidence
UPSCALE_MIN_H     = 60     # upscale crop 2× when its height is below this (px)
NMS_IOU_THRESH    = 0.3    # IoU threshold for NMS deduplication
GAP_MULTIPLIER    = 2.0    # horizontal gap limit = GAP_MULTIPLIER ×␣
 ↪median_char_width
MIN_AREA_FRACTION = 0.0001 # drop merged boxes smaller than this fraction of␣
 ↪page area
```

[6]:
```
import matplotlib.pyplot as plt
from src.textDetection import detect_text_lines

#   Run detection
merged_boxes = detect_text_lines(
    image, layout, ocr,
    min_area_fraction=MIN_AREA_FRACTION,
)

#   Visualise
vis = cv2.cvtColor(image, cv2.COLOR_BGR2RGB).copy()

# Layout region boundaries - RED
for region in layout:
    if region['label'] not in TEXT_LABELS:
        continue
    rx1, ry1, rx2, ry2 = [int(c) for c in region['bbox']]
    cv2.rectangle(vis, (rx1, ry1), (rx2, ry2), (255, 0, 0), 3)
    cv2.putText(
        vis, region['label'],
        (rx1, max(ry1 - 8, 0)),
        cv2.FONT_HERSHEY_SIMPLEX, 0.7, (255, 0, 0), 2
    )

# Merged line boxes - GREEN
for box in merged_boxes:
    pts = np.array(box, np.int32).reshape((-1, 1, 2))
    cv2.polylines(vis, [pts], isClosed=True, color=(0, 255, 0), thickness=2)
```

```
text_region_count = sum(1 for r in layout if r['label'] in TEXT_LABELS)

plt.figure(figsize=(12, 16))
plt.imshow(vis)
plt.axis('off')
plt.title(
    f"Layout-aware Line Detection  |  {text_region_count} text regions  "
    f"|  {len(merged_boxes)} line boxes",
    fontsize=14
)
plt.show()
del vis
```

```
  [text]  bbox=[704, 1425, 2403, 2173]  scale=1.0x  →  18 raw boxes (after score
filter)
  [text]  bbox=[716, 2252, 2451, 3572]  scale=1.0x  →  18 raw boxes (after score
filter)
  [paragraph_title]  bbox=[1122, 1181, 1953, 1399]  scale=1.0x  →  3 raw boxes
(after score filter)

Raw boxes before NMS : 39
Raw boxes after NMS  : 39
Boxes after page-relative filter : 18  (removed 2 ← < 0.010% of page area)
Line-level boxes     : 18
```

Layout-aware Line Detection | 3 text regions | 18 line boxes



POR

DOÑA CATALINA DE Laçarraga y Zarate, viuda de Marco Antonio de Caycedo, y el ſeñor Fiſcal.

CON

Don Antonio del Barco y Recalde, Cauallero de la Orden de Santiago.

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A ca

3 Dataset Generation

[3]:
```
from paddleocr import PaddleOCR
from src.dataUtils import build_ocr_dataset
ocr = PaddleOCR(
    # Specify the exact official model names
    text_detection_model_name='PP-OCRv5_server_det',
    text_recognition_model_name='PP-OCRv5_server_rec',

    # Optional: Turn off the extra document processing modules to save RAM/Time
    use_doc_orientation_classify=False,
    use_doc_unwarping=False,
    use_textline_orientation=False,
    lang='sp'
)
```

```
/tmp/ipykernel_2737379/295365601.py:2: UserWarning: `lang` and `ocr_version`
will be ignored when model names or model directories are not `None`.
  ocr = PaddleOCR(
Creating model: ('PP-OCRv5_server_det', None)
Model files already exist. Using cached files. To redownload, please delete
the directory manually: `/home/sarthak/.paddlex/official_models/PP-
OCRv5_server_det`.
Creating model: ('PP-OCRv5_server_rec', None)
Model files already exist. Using cached files. To redownload, please delete
the directory manually: `/home/sarthak/.paddlex/official_models/PP-
OCRv5_server_rec`.
```

[ ]:
```
build_ocr_dataset(
    image_dir      = "data/3.processed/PORCONES.23.5 - 1628",
    transcript_dir = "data/transcripts/PrintedPages/PORCONES.23.5 - 1628␣
 ↪transcription",
    output_dir     = "data/dataset",
    ocr            = ocr,
)
```

4 3. Text Recognition

4.1 Using CNN-LSTM OCR Model

[1]:
```
# ============================
# CONFIGURATION
# ============================
import torch
import os
```

```python
DATASET_DIR = "data/dataset"

BATCH_SIZE          = 8
NUM_EPOCHS          = 150
LR                  = 3e-4
WEIGHT_DECAY        = 1e-4
EARLY_STOP_PATIENCE = 20

IMG_HEIGHT  = 64          # standard CRNN height; divisible by 8

LSTM_HIDDEN = 256
LSTM_LAYERS = 3

DROPOUT = 0.3

#   Checkpoint directories
CHECKPOINT_DIR  = "checkpoints"
CRNN_WEIGHT_DIR = os.path.join(CHECKPOINT_DIR, "crnn", "weights")
CRNN_PLOT_DIR   = os.path.join(CHECKPOINT_DIR, "crnn", "plots")
TROCR_CKPT_DIR  = os.path.join(CHECKPOINT_DIR, "trocr")

for _d in [CRNN_WEIGHT_DIR, CRNN_PLOT_DIR, TROCR_CKPT_DIR]:
    os.makedirs(_d, exist_ok=True)

MODEL_SAVE_PATH = os.path.join(CRNN_WEIGHT_DIR, "best_crnn.pth")

#   Hardware
DEVICE  = "cuda" if torch.cuda.is_available() else "cpu"
USE_AMP = torch.cuda.is_available()   # fp16 mixed-precision when GPU present

print(f"Using device : {DEVICE}")
print(f"AMP enabled  : {USE_AMP}")
print(f"Checkpoints  : {CHECKPOINT_DIR}/")
```

```
Using device : cuda
AMP enabled  : True
Checkpoints  : checkpoints/
```

[2]:
```python
from src.modelutils.dataset import build_vocab
```

[3]:
```python
from src.modelutils.dataset import LineDataset, resize_keep_ratio
```

[4]:
```python
from src.modelutils.dataset import collate_fn
```

[5]:
```python
import random
from torch.utils.data import DataLoader, Subset
```

```
#   Vocabulary
vocab, char2idx, idx2char = build_vocab(DATASET_DIR)
print(f"Vocab size: {len(vocab)}")

#   Deterministic train / val split
_base_ds = LineDataset(DATASET_DIR, char2idx, img_height=IMG_HEIGHT,␣
 ↪augment=False)
n         = len(_base_ds)
indices   = list(range(n))
random.seed(42)
random.shuffle(indices)

train_size = int(0.9 * n)
train_idx, val_idx = indices[:train_size], indices[train_size:]

train_ds = LineDataset(DATASET_DIR, char2idx, img_height=IMG_HEIGHT,␣
 ↪augment=True)
val_ds   = LineDataset(DATASET_DIR, char2idx, img_height=IMG_HEIGHT,␣
 ↪augment=False)

train_dataset = Subset(train_ds, train_idx)
val_dataset   = Subset(val_ds,   val_idx)
dataset       = _base_ds   # kept for visualisation

print(f"Train: {len(train_dataset)}  |  Val: {len(val_dataset)}")

#   DataLoaders
train_loader = DataLoader(
    train_dataset, batch_size=BATCH_SIZE, shuffle=True,
    collate_fn=collate_fn, num_workers=4, pin_memory=(DEVICE == "cuda"),
)
val_loader = DataLoader(
    val_dataset, batch_size=BATCH_SIZE, shuffle=False,
    collate_fn=collate_fn, num_workers=4, pin_memory=(DEVICE == "cuda"),
)
```

```
Vocab size: 79
LineDataset: 775 samples  (augment=False)
LineDataset: 775 samples  (augment=True)
LineDataset: 775 samples  (augment=False)
Train: 697  |  Val: 78
```

[6]:
```
for imgs, labels, label_lengths, labels_str in train_loader:

    print("Images:", imgs.shape)
    print("Labels:", labels.shape)
    print("Label lengths:", label_lengths)
```

```
    break
```

```
Images: torch.Size([8, 1, 64, 1635])
Labels: torch.Size([599])
Label lengths: tensor([ 51,  56,  94, 100,  53,  51,  95,  99])
```

[7]:
```
import torch
from src.models.crnn import CRNN

#   Sanity check
_dummy   = torch.zeros(2, 1, IMG_HEIGHT, 256)
_m       = CRNN(vocab_size=10, lstm_hidden=LSTM_HIDDEN,␣
 ↪lstm_layers=LSTM_LAYERS, dropout=DROPOUT)
_out     = _m(_dummy)
_n_param = sum(p.numel() for p in _m.parameters())
print(f"CRNN output : {_out.shape}   params : {_n_param:,}")
del _dummy, _m, _out
```

```
CRNN output : torch.Size([2, 64, 10])   params : 8,166,346
```

[8]:
```
from src.evals.metrics import ctc_greedy_decode, compute_cer, compute_wer
```

[9]:
```
from src.modelutils.train import train_epoch, validate, EarlyStopping
```

[10]:
```
import torch.nn as nn
import torch.optim as optim
import json
from tqdm import tqdm

#   Model, optimiser, loss
model     = CRNN(len(vocab), lstm_hidden=LSTM_HIDDEN, lstm_layers=LSTM_LAYERS,␣
 ↪dropout=DROPOUT).to(DEVICE)
criterion = nn.CTCLoss(blank=0, zero_infinity=True)
optimizer = optim.AdamW(model.parameters(), lr=LR, weight_decay=WEIGHT_DECAY)

scheduler = optim.lr_scheduler.OneCycleLR(
    optimizer,
    max_lr           = LR,
    steps_per_epoch  = len(train_loader),
    epochs           = NUM_EPOCHS,
    pct_start        = 0.05,
    anneal_strategy  = "cos",
    div_factor       = 25,
    final_div_factor = 1e4,
)

scaler = torch.amp.GradScaler("cuda", enabled=USE_AMP)
```

```
#   Early stopping & history
early_stop    = EarlyStopping(EARLY_STOP_PATIENCE)
best_val_loss = float("inf")
train_losses, val_losses = [], []
train_cers,   val_cers   = [], []
train_wers,   val_wers   = [], []

n_params = sum(p.numel() for p in model.parameters())
print(f"Device : {DEVICE}  |  AMP: {USE_AMP}  |  Vocab: {len(vocab)}  |  Params:
 ↪ {n_params:,}")

#   Training loop
epoch_bar = tqdm(range(1, NUM_EPOCHS + 1), desc="Training", unit="ep",␣
 ↪dynamic_ncols=True)

for epoch in epoch_bar:

    t_loss, t_cer, t_wer = train_epoch(
        model, train_loader, optimizer, criterion, scaler,
        idx2char=idx2char, device=DEVICE, use_amp=USE_AMP, scheduler=scheduler,
    )
    v_loss, v_cer, v_wer = validate(
        model, val_loader, criterion,
        idx2char=idx2char, device=DEVICE, use_amp=USE_AMP,
    )

    train_losses.append(t_loss); val_losses.append(v_loss)
    train_cers.append(t_cer);    val_cers.append(v_cer)
    train_wers.append(t_wer);    val_wers.append(v_wer)

    lr_now = optimizer.param_groups[0]["lr"]
    saved  = ""

    if v_loss < best_val_loss:
        best_val_loss = v_loss
        saved = "  "
        torch.save({
            "epoch":                epoch,
            "model_state_dict":     model.state_dict(),
            "optimizer_state_dict": optimizer.state_dict(),
            "scaler_state_dict":    scaler.state_dict(),
            "val_loss": v_loss, "val_cer": v_cer, "val_wer": v_wer,
            "train_loss": t_loss,
            "vocab": vocab, "char2idx": char2idx, "idx2char": idx2char,
            "img_height": IMG_HEIGHT, "lstm_hidden": LSTM_HIDDEN, "lstm_layers":
 ↪ LSTM_LAYERS,
```

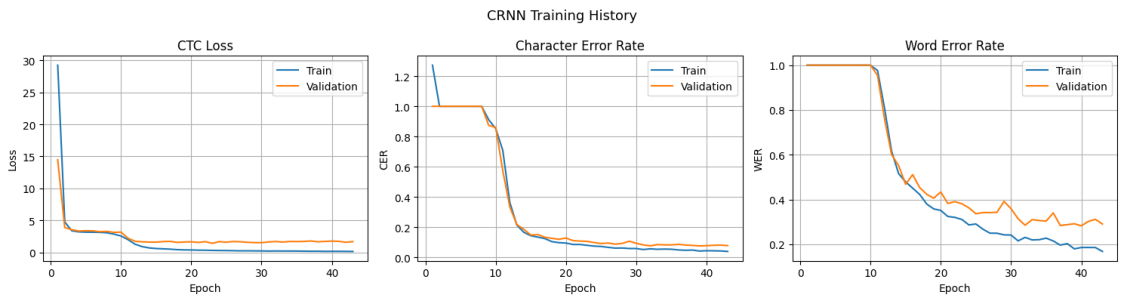
```
        }, MODEL_SAVE_PATH)
        with open(os.path.join(CRNN_WEIGHT_DIR, "history.json"), "w") as _f:
            json.dump({
                "train_losses": train_losses, "val_losses": val_losses,
                "train_cers": train_cers,     "val_cers":   val_cers,
                "train_wers": train_wers,     "val_wers":   val_wers,
            }, _f)

    epoch_bar.set_postfix(
        tL=f"{t_loss:.3f}", tCER=f"{t_cer:.3f}", tWER=f"{t_wer:.3f}",
        vL=f"{v_loss:.3f}", vCER=f"{v_cer:.3f}", vWER=f"{v_wer:.3f}",
        lr=f"{lr_now:.1e}", best=f"{best_val_loss:.3f}", saved=saved,
    )

    if DEVICE == "cuda":
        torch.cuda.empty_cache()

    if early_stop.step(v_loss):
        epoch_bar.write(f"\nEarly stopping at epoch {epoch}.")
        break

epoch_bar.close()
print(f"\nTraining complete.  Best val loss : {best_val_loss:.4f}")
print(f"Checkpoint → {MODEL_SAVE_PATH}")
```

```
Device : cuda  |  AMP: True  |  Vocab: 79  |  Params: 8,201,743

Training:   0%|          | 0/150 [00:00<?, ?ep/s]/home/sarthak/Documents/RenAIss
anceExperimental/src/modelutils/train.py:84: UserWarning: Detected call of
`lr_scheduler.step()` before `optimizer.step()`. In PyTorch 1.1.0 and later, you
should call them in the opposite order: `optimizer.step()` before
`lr_scheduler.step()`.  Failure to do this will result in PyTorch skipping the
first value of the learning rate schedule. See more details at
https://pytorch.org/docs/stable/optim.html#how-to-adjust-learning-rate
  scheduler.step()
Training:  28%|         | 42/150 [16:54<43:29, 24.17s/ep, best=1.410,
lr=2.6e-04, saved=, tCER=0.039, tL=0.137, tWER=0.168, vCER=0.078, vL=1.675,
vWER=0.290]


Early stopping at epoch 43.

Training complete.  Best val loss : 1.4105
Checkpoint → checkpoints/crnn/weights/best_crnn.pth
```

[11]:
```
import os
from src.plots.training import plot_training_history
```

```
_plot_path = os.path.join(CRNN_PLOT_DIR, "training_history.png")
plot_training_history(
    train_losses, val_losses,
    train_cers,   val_cers,
    train_wers,   val_wers,
    save_path=_plot_path,
    title="CRNN Training History",
)
```

```
Plot saved → checkpoints/crnn/plots/training_history.png
```



[13]:
```
import torch
import os
from src.plots.predictions import plot_sample_predictions

# Load best checkpoint into the model
checkpoint = torch.load(MODEL_SAVE_PATH, map_location=DEVICE)
model.load_state_dict(checkpoint["model_state_dict"])

_pred_plot = os.path.join(CRNN_PLOT_DIR, "sample_predictions.png")
plot_sample_predictions(
    model,
    val_dataset,
    idx2char,
    checkpoint   = checkpoint,
    img_height   = IMG_HEIGHT,
    device       = DEVICE,
    num_samples  = 6,
    save_path    = _pred_plot,
)
```

```
Plot saved → checkpoints/crnn/plots/sample_predictions.png
```

5 Using TrOCR

[14]:
```
import os
from dotenv import load_dotenv

load_dotenv()  # loads variables from .env file

hf_token = os.getenv("HF_TOKEN")
print("Token loaded:", hf_token is not None)

from transformers import TrOCRProcessor, VisionEncoderDecoderModel
import torch

device = "cuda" if torch.cuda.is_available() else "cpu"
print("Using device:", device)
```

```
Token loaded: True
Using device: cuda
```

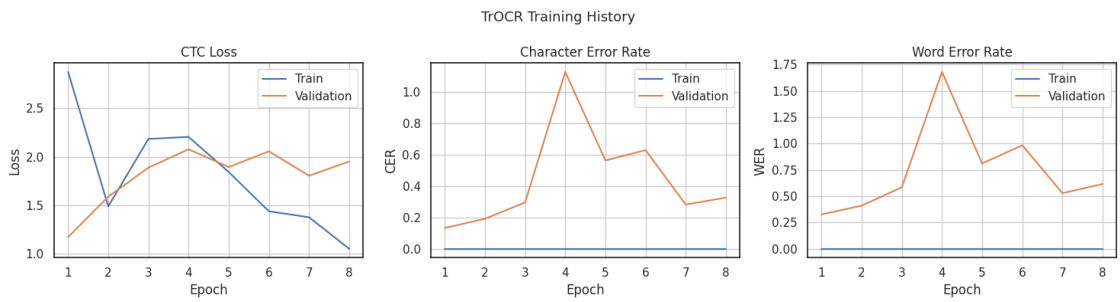
[15]:
```python
import os, torch

#   Hardware
DEVICE        = "cuda" if torch.cuda.is_available() else "cpu"
USE_AMP_TROCR = (DEVICE == "cuda")

#   Paths
DATASET_DIR  = "data/dataset"
TROCR_CKPT_DIR = os.path.join("checkpoints", "trocr")
os.makedirs(TROCR_CKPT_DIR, exist_ok=True)

#   Hyperparameters
TROCR_BASE_MODEL  = "microsoft/trocr-base-printed"
TROCR_EPOCHS      = 30
TROCR_BATCH_SIZE  = 4
TROCR_LR          = 5e-5
TROCR_MAX_LEN     = 128
TROCR_ES_PATIENCE = 7

print(f"TrOCR config ready  |  device: {DEVICE}  |  AMP: {USE_AMP_TROCR}")
print(f"Save dir: {TROCR_CKPT_DIR}")
```

```
TrOCR config ready  |  device: cuda  |  AMP: True
Save dir: checkpoints/trocr
```

[16]:
```python
from src.modelutils.train import train_trocr

trocr_history = train_trocr(
    dataset_dir    = DATASET_DIR,
    save_dir       = TROCR_CKPT_DIR,
    base_model     = TROCR_BASE_MODEL,
    epochs         = TROCR_EPOCHS,
    batch_size     = TROCR_BATCH_SIZE,
    lr             = TROCR_LR,
    max_target_len = TROCR_MAX_LEN,
    es_patience    = TROCR_ES_PATIENCE,
    device         = DEVICE,
    use_amp        = USE_AMP_TROCR,
    plot_save_path = os.path.join(TROCR_CKPT_DIR, "trocr_training_history.png"),
)
print(f"Best val loss: {trocr_history['best_loss']:.4f}")
```

```
Loading  microsoft/trocr-base-printed …

Loading weights:   0%|          | 0/478 [00:00<?, ?it/s]

TrOCRLineDataset: 775 samples
TrOCR params: 333,921,792  |  device: cuda  |  AMP: True

TrOCR Fine-tune:   0%|          | 0/30 [00:00<?, ?ep/s]
```

```
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]

TrOCR Fine-tune:  23%|         | 7/30 [12:14<40:12, 104.90s/ep, best=1.1700,
saved=, tL=1.0488, vCER=0.326, vL=1.9511, vWER=0.615]


Early stopping at epoch 8.

TrOCR fine-tuning complete.  Best val loss: 1.1700
Model saved → checkpoints/trocr
Plot saved → checkpoints/trocr/trocr_training_history.png
```




```
Best val loss: 1.1700
```

6 4. Inference

[1]:
```python
import os, torch

DEVICE         = "cuda" if torch.cuda.is_available() else "cpu"

CHECKPOINT_DIR = "checkpoints"
MODEL_SAVE_PATH = os.path.join(CHECKPOINT_DIR, "crnn", "weights", "best_crnn.
 ↪pth")
TROCR_CKPT_DIR  = os.path.join(CHECKPOINT_DIR, "trocr")

print(f"Inference device   : {DEVICE}")
print(f"CRNN checkpoint    : {MODEL_SAVE_PATH}")
print(f"TrOCR checkpoint   : {TROCR_CKPT_DIR}")
```

```
Inference device   : cuda
CRNN checkpoint    : checkpoints/crnn/weights/best_crnn.pth
TrOCR checkpoint   : checkpoints/trocr
```

6.1 CRNN: Full-Page Inference

Takes a full-page image path and runs the same **layout-aware** two-stage detection used in Section 2: 1. `PP-DocLayout_plus-L` detects text regions (paragraphs, titles, headers) — freed after use. 2. `PP-OCRv5_server_det` runs line detection **per region** with NMS, score filtering and small-region

upscaling — freed before the recognition model loads. 3. The trained **CRNN** model recognises each cropped line — freed after inference. 4. Visualisation (no model in memory).

[2]:
```
from src.inference.crnn_infer import crnn_infer_page

_PAGE = "data/3.processed/PORCONES.23.5 - 1628/page1.png"   # ← change as needed

crnn_lines = crnn_infer_page(
    image_path      = _PAGE,
    checkpoint_path = MODEL_SAVE_PATH,
    device          = DEVICE,
    visualize       = True,
)

print(f"\n{'='*60}")
print(f"CRNN Full-Page Transcript  ({len(crnn_lines)} lines)")
print(f"{'='*60}")
for _i, _line in enumerate(crnn_lines, 1):
    print(f"[{_i:3d}] {_line}")
```

```
[1/4] Running layout detection …

/home/sarthak/miniconda3/envs/RenaissanceDetect/lib/python3.10/site-
packages/requests/__init__.py:113: RequestsDependencyWarning: urllib3 (2.6.3) or
chardet (6.0.0.post1)/charset_normalizer (3.4.4) doesn't match a supported
version!
  warnings.warn(
Checking connectivity to the model hosters, this may take a while. To

bypass this check, set `PADDLE_PDX_DISABLE_MODEL_SOURCE_CHECK` to `True`.
Creating model: ('PP-DocLayout_plus-L', None)
Model files already exist. Using cached files. To redownload, please delete

the directory manually: `/home/sarthak/.paddlex/official_models/PP-

DocLayout_plus-L`.
Creating model: ('PP-LCNet_x1_0_textline_ori', None)
Model files already exist. Using cached files. To redownload, please delete

the directory manually: `/home/sarthak/.paddlex/official_models/PP-

LCNet_x1_0_textline_ori`.
Creating model: ('PP-OCRv5_server_det', None)
Model files already exist. Using cached files. To redownload, please delete

the directory manually: `/home/sarthak/.paddlex/official_models/PP-

OCRv5_server_det`.
Creating model: ('PP-OCRv5_server_rec', None)
```

```
Model files already exist. Using cached files. To redownload, please delete
the directory manually: `/home/sarthak/.paddlex/official_models/PP-
OCRv5_server_rec`.
Creating model: ('PP-Chart2Table', None)
Model files already exist. Using cached files. To redownload, please delete
the directory manually: `/home/sarthak/.paddlex/official_models/PP-
Chart2Table`.
Special tokens have been added in the vocabulary, make sure the associated
word embeddings are fine-tuned or trained.
Loading configuration file /home/sarthak/.paddlex/official_models/PP-
Chart2Table/config.json
Loading weights file /home/sarthak/.paddlex/official_models/PP-
Chart2Table/model_state.pdparams
Loaded weights file from disk, setting weights to model.
All model checkpoint weights were used when initializing
PPChart2TableInference.

All the weights of PPChart2TableInference were initialized from the model
checkpoint at /home/sarthak/.paddlex/official_models/PP-Chart2Table.
If your task is similar to the task the model of the checkpoint was trained on,
you can already use PPChart2TableInference for predictions without further
training.
Loading configuration file /home/sarthak/.paddlex/official_models/PP-
Chart2Table/generation_config.json
/home/sarthak/Documents/RenAIssanceExperimental/src/inference/crnn_infer.py:126:
UserWarning: `lang` and `ocr_version` will be ignored when model names or model
directories are not `None`.
  _ocr = PaddleOCR(
Creating model: ('PP-OCRv5_server_det', None)
Model files already exist. Using cached files. To redownload, please delete
the directory manually: `/home/sarthak/.paddlex/official_models/PP-
OCRv5_server_det`.

      Found 3 text regions.
[2/4] Running layout-aware line detection …
Creating model: ('PP-OCRv5_server_rec', None)
```

```
Model files already exist. Using cached files. To redownload, please delete
the directory manually: `/home/sarthak/.paddlex/official_models/PP-
OCRv5_server_rec`.
  [text]  bbox=[704, 1425, 2403, 2173]  scale=1.0x  →  18 raw boxes (after score
filter)
  [text]  bbox=[716, 2252, 2451, 3572]  scale=1.0x  →  18 raw boxes (after score
filter)
  [paragraph_title]  bbox=[1122, 1181, 1953, 1399]  scale=1.0x  →  3 raw boxes
(after score filter)

Raw boxes before NMS : 39
Raw boxes after NMS  : 39
Boxes after page-relative filter : 18  (removed 2 ← < 0.010% of page area)
Line-level boxes     : 18
      Detected 18 text lines.
[3/4] Running CRNN recognition …
      Recognised 18 lines.
[4/4] Visualising …
```

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L 23 nº 5

POR

DOÑA CATALINA DE Laçarraga y Zarate, viuda de Marco Antonio de Caycedo, y el ſeñor Fiſcal.

CON

Don Antonio del Barco y Recalde, Cauallero de la Orden de Santiago.

PRetende doña Catalina, que dõ Antonio ha de ſer condenado a q̃ auiendoſele quitado el habito, e inſignias de la Orden, ſea entregado a la juſticia ſeglar, para que ſe execute en èl la pena de muerte, por los delitos q̃ cometiò de auer con malos medios, y tercerias, y promeſſas, y alagos engañado a D. Frãciſ-

A ca

==========

```
CRNN Full-Page Transcript  (18 lines)
============================================================
[  1] POR
[  2] DONA CATAIINA DE
[  3] Laxarraga y zarate viuda de Marco
[  4] Antoniode Caycedo, y el seior
[  5] iscal.
[  6] ON
[  7] Don Antoniodel Barco y Recalde,
[  8] Cavallero de la Orien ae
[  9] Santiago.
[ 10] Petende doña Catalina que do An
[ 11] tonio ha de ser condenado a q avien
[ 12] dosele quitado el habito, e insignias
[ 13] de la Orden, sea entregado a la justi
[ 14] cia seglar, para que se execute en ei
[ 15] la pena de muerte, por los delitos q
[ 16] cometio de aver conmalos medios,
[ 17] y tercerias, y promessas, y alagos engañado aD. Fracis
[ 18] na
```

6.2 TrOCR: Full-Page Inference

Takes a full-page image path and runs the same **layout-aware** two-stage detection used in Section 2: 1. `PP-DocLayout_plus-L` detects text regions (paragraphs, titles, headers) — freed after use. 2. `PP-OCRv5_server_det` runs line detection **per region** with NMS, score filtering and small-region upscaling — freed before the recognition model loads. 3. The fine-tuned **TrOCR** model recognises each cropped line — freed after inference. 4. Visualisation (no model in memory).

[4]:
```python
from src.inference.trocr_infer import trocr_infer_page

_PAGE = "data/3.processed/PORCONES.23.5 - 1628/page1.png"   # ← change as needed

trocr_lines = trocr_infer_page(
    image_path = _PAGE,
    model_dir  = TROCR_CKPT_DIR,
    device     = DEVICE,
    visualize  = True,
)

print(f"\n{'='*60}")
print(f"TrOCR Full-Page Transcript  ({len(trocr_lines)} lines)")
print(f"{'='*60}")
for _i, _line in enumerate(trocr_lines, 1):
    print(f"[{_i:3d}] {_line}")
```

```
Creating model: ('PP-DocLayout_plus-L', None)
Model files already exist. Using cached files. To redownload, please delete
the directory manually: `/home/sarthak/.paddlex/official_models/PP-
DocLayout_plus-L`.
[1/4] Running layout detection …
Creating model: ('PP-LCNet_x1_0_textline_ori', None)
Model files already exist. Using cached files. To redownload, please delete
the directory manually: `/home/sarthak/.paddlex/official_models/PP-
LCNet_x1_0_textline_ori`.
Creating model: ('PP-OCRv5_server_det', None)
Model files already exist. Using cached files. To redownload, please delete
the directory manually: `/home/sarthak/.paddlex/official_models/PP-
OCRv5_server_det`.
Creating model: ('PP-OCRv5_server_rec', None)
Model files already exist. Using cached files. To redownload, please delete
the directory manually: `/home/sarthak/.paddlex/official_models/PP-
OCRv5_server_rec`.
Creating model: ('PP-Chart2Table', None)
Model files already exist. Using cached files. To redownload, please delete
the directory manually: `/home/sarthak/.paddlex/official_models/PP-
Chart2Table`.
Special tokens have been added in the vocabulary, make sure the associated
word embeddings are fine-tuned or trained.
Loading configuration file /home/sarthak/.paddlex/official_models/PP-
Chart2Table/config.json
Loading weights file /home/sarthak/.paddlex/official_models/PP-
Chart2Table/model_state.pdparams
Loaded weights file from disk, setting weights to model.
All model checkpoint weights were used when initializing
PPChart2TableInference.

All the weights of PPChart2TableInference were initialized from the model
checkpoint at /home/sarthak/.paddlex/official_models/PP-Chart2Table.
If your task is similar to the task the model of the checkpoint was trained on,
you can already use PPChart2TableInference for predictions without further
training.
```

```
Loading configuration file /home/sarthak/.paddlex/official_models/PP-
Chart2Table/generation_config.json
/home/sarthak/Documents/RenAIssanceExperimental/src/inference/trocr_infer.py:95:
UserWarning: `lang` and `ocr_version` will be ignored when model names or model
directories are not `None`.
  _ocr = PaddleOCR(
Creating model: ('PP-OCRv5_server_det', None)
Model files already exist. Using cached files. To redownload, please delete
the directory manually: `/home/sarthak/.paddlex/official_models/PP-
OCRv5_server_det`.

      Found 3 text regions.
[2/4] Running layout-aware line detection …
Creating model: ('PP-OCRv5_server_rec', None)
Model files already exist. Using cached files. To redownload, please delete
the directory manually: `/home/sarthak/.paddlex/official_models/PP-
OCRv5_server_rec`.

  [text]  bbox=[704, 1425, 2403, 2173]  scale=1.0x  →  18 raw boxes (after score
filter)
  [text]  bbox=[716, 2252, 2451, 3572]  scale=1.0x  →  18 raw boxes (after score
filter)
  [paragraph_title]  bbox=[1122, 1181, 1953, 1399]  scale=1.0x  →  3 raw boxes
(after score filter)

Raw boxes before NMS : 39
Raw boxes after NMS  : 39
Boxes after page-relative filter : 18  (removed 2 ← < 0.010% of page area)
Line-level boxes     : 18
      Detected 18 text lines.
[3/4] Loading TrOCR from checkpoints/trocr …
Loading weights:   0%|          | 0/480 [00:00<?, ?it/s]

      Recognised 18 lines.
[4/4] Visualising …
```

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L 23 nº 5

POR

DOÑA CATALINA DE Laçarraga y Zarate, viuda de Marco Antonio de Caycedo, y el ſeñor Fiſcal.

CON

Don Antonio del Barco y Recalde, Cauallero de la Orden de Santiago.

PRetende doña Catalina, que dõ Antonio ha de ſer condenado a q̃ auiendoſele quitado el habito, e inſignias de la Orden, ſea entregado a la juſticia ſeglar, para que ſe execute en èl la pena de muerte, por los delitos q̃ cometiò de auer con malos medios, y tercerias, y promeſſas, y alagos engañado a D. Frãciſ-

A ca

==========

```
TrOCR Full-Page Transcript  (18 lines)
============================================================
[  1] POUR
[  2] DONA CATALINA DE
[  3] Lacarraga y Zarate, viuda de Marco
[  4] Antonio de Caycedo, y el señor
[  5] Fiscal.
[  6] O N
[  7] Don Antonio del Barco y Recalde,
[  8] Cauallero de la Orden de
[  9] SANTAgo.
[ 10] Retende doña Catalina, que do An-
[ 11] tonio ha de ser condenado a q avien
[ 12] dosle quirado el habito, e infignias
[ 13] de la Orden, sea entregado a la justi-
[ 14] cia regular, para que se execute en el
[ 15] Ia pena de muerte, por los delitos q
[ 16] cometio de aver con malos medios,
[ 17] y tercerias, y promesias, y alagos engaiado a D. Fraci-
[ 18] A ~~
```

7 5. LLM Post-Processing

Raw OCR output often contains character-level errors, missing diacritics and broken word boundaries — especially for historical Spanish texts. This section uses the **Google Gemini API** to pass the raw OCR transcript through an expert LLM that:

- Acts as a **Spanish historical document expert** with deep knowledge of early-modern orthography, archaic spelling conventions (v/b, x/j, ç/s), and period vocabulary (legal, religious, administrative).
- **Reconstructs ambiguous or garbled tokens** using linguistic and contextual knowledge — confident completion of words the OCR mangled.
- Restores diacritics, fixes OCR glyph confusions (`rn`→`m`, `0`→`o`, long-f, etc.) and corrects tokenisation errors.
- Preserves the original language, style and line structure — no translation or modernisation.

Workflow: 1. Join the per-line OCR strings into a full-page transcript. 2. Send chunks to Gemini with a detailed expert system prompt. 3. Display a side-by-side diff of raw vs. corrected transcript.

Set `GEMINI_API_KEY` in your `.env` file before running.

[5]:
```python
from src.postprocess.llm import gemini_postprocess_transcript, print_diff
from src.evals.metrics import compute_cer, compute_wer

GEMINI_MODEL = "gemini-3-flash-preview"
print(f"Gemini postprocessor ready  |  model: {GEMINI_MODEL}")
```

```
Gemini postprocessor ready  |  model: gemini-3-flash-preview
```

7.1 5.1 Post-process CRNN Transcript

[6]:
```
try:
    _ = crnn_lines
except NameError:
    raise RuntimeError(
        "crnn_lines is not defined. "
        "Please run the 'CRNN: Full-Page Inference' cell in Section 4 first."
    )

print(f"Post-processing {len(crnn_lines)} CRNN lines with Gemini␣
↪({GEMINI_MODEL}) ...")
crnn_corrected = gemini_postprocess_transcript(crnn_lines, model=GEMINI_MODEL)

print_diff(crnn_lines, crnn_corrected, title="CRNN → Gemini Post-Processed")

crnn_full_transcript = "\n".join(crnn_corrected)
print(f"\n{'='*70}\nFull corrected CRNN transcript:\n{'='*70}")
print(crnn_full_transcript)
```

```
[2026-03-12 09:33:48,831] [    INFO] models.py:4672 - AFC is enabled with max
remote calls: 10.

Post-processing 18 CRNN lines with Gemini (gemini-3-flash-preview) …
    Sending full transcript (18 lines) to Gemini (gemini-3-flash-preview) …

[2026-03-12 09:34:32,616] [    INFO] models.py:4683 - AFC remote call 1 is done.

    Done - 16 / 18 lines corrected.

======================================================================
  CRNN → Gemini Post-Processed
======================================================================
RAW                                                                  CORRECTED
--------------------------------------------------------------------------------
-------------------------------------------------------------
POR                                                                  POR
DONA CATAIINA DE                                                     DOÑA
CATALINA DE
Laxarraga y zarate viuda de Marco                                    Laxarraga
y Zárate viuda de Marco
Antoniode Caycedo, y el seior                                        Antonio de
Caycedo, y el señor
iscal.                                                               Fiscal.

ON                                                                   CON

Don Antoniodel Barco y Recalde,                                      Don
Antonio del Barco y Recalde,
```

```
Cavallero de la Orien ae                                                Cavallero
de la Orden de
Santiago.                                                               Santiago.
Petende doña Catalina que do An                                         Pretende
doña Catalina que don An
tonio ha de ser condenado a q avien                                     tonio ha
de ser condenado á que avién
dosele quitado el habito, e insignias                                   dosele
quitado el hábito, e insignias
de la Orden, sea entregado a la justi                                   de la
Orden, sea entregado á la justi
cia seglar, para que se execute en ei                                   cia
seglar, para que se execute en el
la pena de muerte, por los delitos q                                    la pena de
muerte, por los delitos que
cometio de aver conmalos medios,                                        cometió de
aver con malos medios,
y tercerias, y promessas, y alagos engañado aD. Fracis                  y
tercerías, y promessas, y alagos engañado á D. Francis
na                                                                      ca


16/18 lines changed by LLM post-processing.

======================================================================
Full corrected CRNN transcript:
======================================================================
POR
DOÑA CATALINA DE
Laxarraga y Zárate viuda de Marco
Antonio de Caycedo, y el señor
Fiscal.
CON
Don Antonio del Barco y Recalde,
Cavallero de la Orden de
Santiago.
Pretende doña Catalina que don An
tonio ha de ser condenado á que avién
dosele quitado el hábito, e insignias
de la Orden, sea entregado á la justi
cia seglar, para que se execute en el
la pena de muerte, por los delitos que
cometió de aver con malos medios,
y tercerías, y promessas, y alagos engañado á D. Francis
ca
```

7.2 5.2 Post-process TrOCR Transcript

[7]:
```
try:
    _ = trocr_lines
except NameError:
    raise RuntimeError(
        "trocr_lines is not defined. "
        "Please run the 'TrOCR: Full-Page Inference' cell in Section 4 first."
    )

print(f"Post-processing {len(trocr_lines)} TrOCR lines with Gemini␣
↪({GEMINI_MODEL}) ...")
trocr_corrected = gemini_postprocess_transcript(trocr_lines, model=GEMINI_MODEL)

print_diff(trocr_lines, trocr_corrected, title="TrOCR → Gemini Post-Processed")

trocr_full_transcript = "\n".join(trocr_corrected)
print(f"\n{'='*70}\nFull corrected TrOCR transcript:\n{'='*70}")
print(trocr_full_transcript)
```

```
[2026-03-12 09:34:32,654] [    INFO] models.py:4672 - AFC is enabled with max
remote calls: 10.

Post-processing 18 TrOCR lines with Gemini (gemini-3-flash-preview) …
    Sending full transcript (18 lines) to Gemini (gemini-3-flash-preview) …

[2026-03-12 09:35:17,334] [    INFO] models.py:4683 - AFC remote call 1 is done.

    Done - 14 / 18 lines corrected.

======================================================================
  TrOCR → Gemini Post-Processed
======================================================================
RAW                                                                  CORRECTED
--------------------------------------------------------------------------------
------------------------------------------------------------
POUR                                                                 POR

DONA CATALINA DE                                                     DOÑA
CATALINA DE
Lacarraga y Zarate, viuda de Marco                                   Luzarraga
y Zárate, viuda de Marco
Antonio de Caycedo, y el señor                                       Antonio de
Caycedo, y el señor
Fiscal.                                                              Fiscal.
O N                                                                  C O N

Don Antonio del Barco y Recalde,                                     Don
Antonio del Barco y Recalde,
```

Cauallero de la Orden de	Cauallero
de la Orden de	
SANTAgo.	SANTIAGO.
Retende doña Catalina, que do An-	Pretende
doña Catalina, que don An-	
tonio ha de ser condenado a q avien	tonio ha
de ser condenado á que avién-	
dosle quirado el habito, e infignias	dosle
quitado el hábito, é insignias	
de la Orden, sea entregado a la justi-	de la
Orden, sea entregado á la justi-	
cia regular, para que se execute en el	cia
secular, para que se execute en él	
Ia pena de muerte, por los delitos q	la pena de
muerte, por los delitos que	
cometio de aver con malos medios,	cometió de
aver con malos medios,	
y tercerias, y promesias, y alagos engaiado a D. Fraci-	y
tercerías, y promessas, y alagos engañado á D. Franci-	
A ~~	sca.

14/18 lines changed by LLM post-processing.

==========
Full corrected TrOCR transcript:
==========

POR
DOÑA CATALINA DE
Luzarraga y Zárate, viuda de Marco
Antonio de Caycedo, y el señor
Fiscal.
C O N
Don Antonio del Barco y Recalde,
Cauallero de la Orden de
SANTIAGO.
Pretende doña Catalina, que don An-
tonio ha de ser condenado á que avién-
dosle quitado el hábito, é insignias
de la Orden, sea entregado á la justi-
cia secular, para que se execute en él
la pena de muerte, por los delitos que
cometió de aver con malos medios,
y tercerías, y promessas, y alagos engañado á D. Franci-
sca.

[11]:
```
from src.evals.metrics import compute_cer, compute_wer

# Assuming you have the ground truth transcript for the page
ground_truth_path = "data/transcripts/PrintedPages/PORCONES.23.5 - 1628␣
↪transcription/page1.txt"
ground_truth = open(ground_truth_path, "r", encoding="utf-8").read().strip()

# Before post-processing
cnn_lstm_cer_before = compute_cer("\n".join(crnn_lines), ground_truth)
cnn_lstm_wer_before = compute_wer("\n".join(crnn_lines), ground_truth)
trocr_cer_before = compute_cer("\n".join(trocr_lines), ground_truth)
trocr_wer_before = compute_wer("\n".join(trocr_lines), ground_truth)
print(f"Before Gemini post-processing:")
print(f"CNN-LSTM CER: {cnn_lstm_cer_before:.3f}  |  WER: {cnn_lstm_wer_before:.
↪3f}")
print(f"TrOCR    CER: {trocr_cer_before:.3f}  |  WER: {trocr_wer_before:.3f}")

# After Gemini post-processing
cnn_lstm_cer = compute_cer(crnn_full_transcript, ground_truth)
cnn_lstm_wer = compute_wer(crnn_full_transcript, ground_truth)
trocr_cer = compute_cer(trocr_full_transcript, ground_truth)
trocr_wer = compute_wer(trocr_full_transcript, ground_truth)
print(f"\nAfter Gemini post-processing:")
print(f"CRNN  + Gemini CER: {cnn_lstm_cer:.3f}  |  WER: {cnn_lstm_wer:.3f}")
print(f"TrOCR + Gemini CER: {trocr_cer:.3f}  |  WER: {trocr_wer:.3f}")
```

```
Before Gemini post-processing:
CNN-LSTM CER: 0.843  |  WER: 1.028
TrOCR    CER: 0.955  |  WER: 1.176

After Gemini post-processing:
CRNN  + Gemini CER: 0.870  |  WER: 1.069
TrOCR + Gemini CER: 0.658  |  WER: 0.809
```